DAXALGO RESEARCH // ARCHITECTURE REVISION

CHMT-Q revision: signal first, policy second

Failure evidence and gated Transformer-DQN architecture | OKX BTC-USDT | July 2026

DECISION

DO NOT DEPLOY. DO NOT ALLOCATE CAPITAL.

The old checkpoint is not tradable. DQN is proposed only after supervised return-signal gates pass.

+2.26%	+0.01%	0	-68.1%
aggregate pinball	**H5 POC median MAE**	**primary-policy trades**	**base-cost return**
vs published marginal baseline	effectively tied with baseline	all intervals crossed zero	median long/flat proxy

WHAT FAILED

Objective

Five footprint distributions, not return, price, action, or net PnL.

Signal

H5 close-direction MCC 0.0475; POC Spearman 0.078.

Economics

0.45 bps gross round-trip edge versus 24 bps modeled cost.

Evidence

18 train days, 2 selection days, 4 forward days, one seed and fold.

Revision status: diagnosis complete; replacement model is not yet trained or validated.

01 // ROOT-CAUSE MAP

The failure is a chain, not one bad hyperparameter

Every link is observed in the authorized Jun 1-24 evidence; causal strength is labeled explicitly.

1 WRONG ENDPOINT

Footprint pinball

No direct return target

→

2 WEAK MEDIAN

H5 POC rho 0.075

Std ratio 0.151

→

3 WIDE RISK

$190.47 interval

100% crosses zero

→

4 ABSTENTION

0 primary trades

Correct under uncertainty

→

5 COST FAILURE

0.45 vs 24 bps

-68.1% base cost

Ranked evidence

PROVEN	Objective does not optimize executable return.
PROVEN	Gross edge is 53.9x below the one-way base-cost hurdle.
PROVEN	Three POC heads are redundant and presence is constant.
PROVEN	Every update was gradient-clipped; schedules stopped at 36%.
PROVEN	Sparse L1 changes lose magnitude at scaler boundaries.
LIKELY	Selection optimism, regime shift, and overfit coexist.

What did not fail

- Checkpoint and source hashes match.
- Scaler is bound to training IDs.
- Horizon alignment reconstructs exactly.
- No NaN or attention-logit runaway.
- No evidence of a compiler off-by-one.

Conclusion: the artifact is internally consistent, but consistency cannot repair a target, sample, and execution design that does not identify tradable edge.

The headline score hides the target that matters

Positive means model improvement over the published empirical marginal; negative means degradation.

3.46%

objective weight

H5 total-POC quantile cell

+2.26%

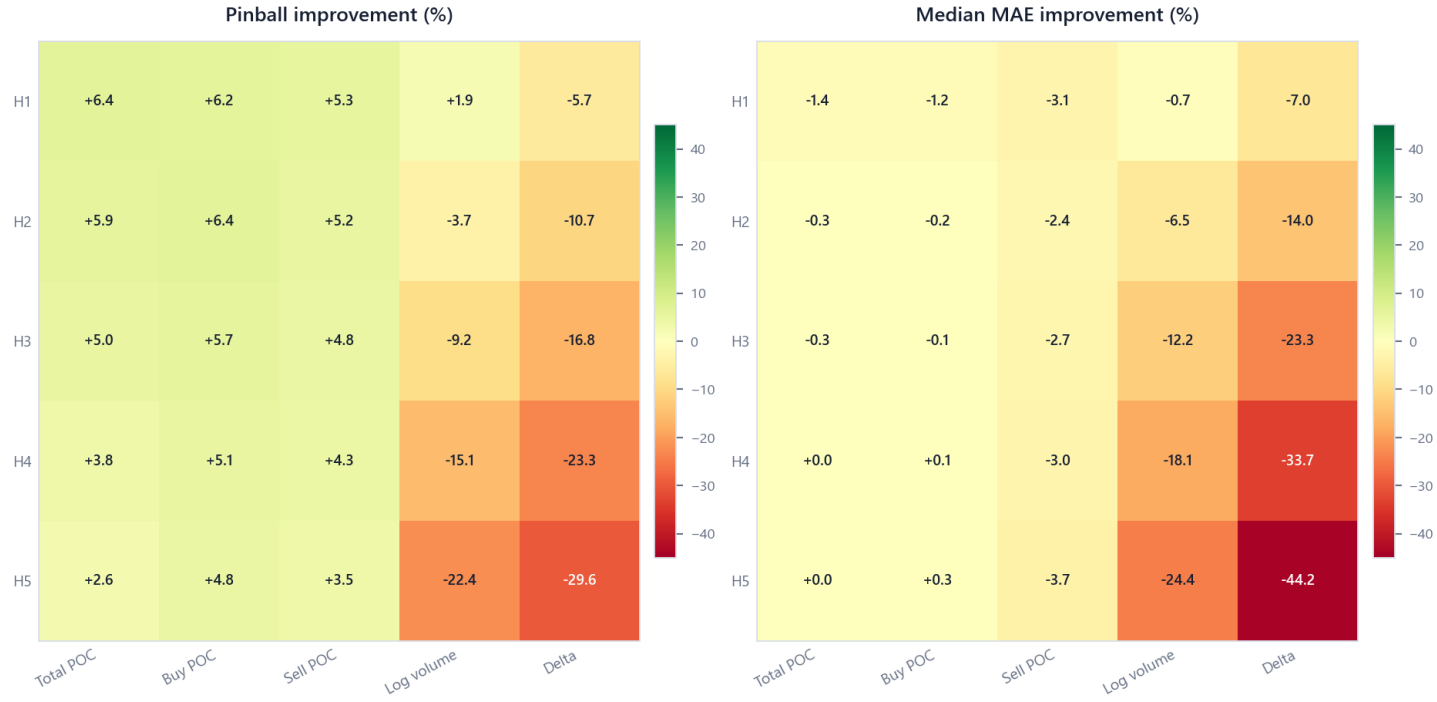
aggregate pinball

all horizons and targets

75%

forecast weight

three highly correlated POC heads

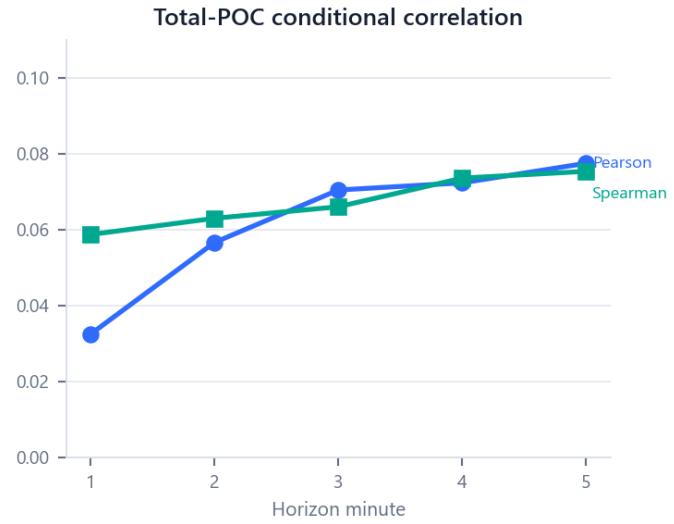


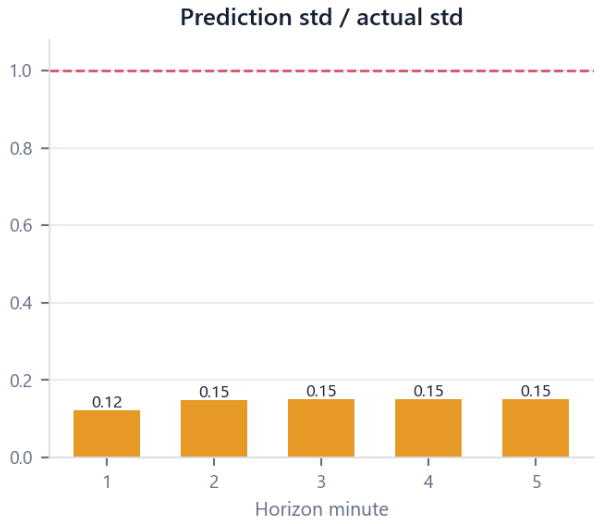
H5 result: total POC pinball +2.62% and median MAE +0.01%; log-volume pinball -22.43% / MAE -24.44%; delta pinball -29.57% / MAE -44.18%.

Model parameters: 2,089,554 | target count: 5 x 5 x 3 quantiles

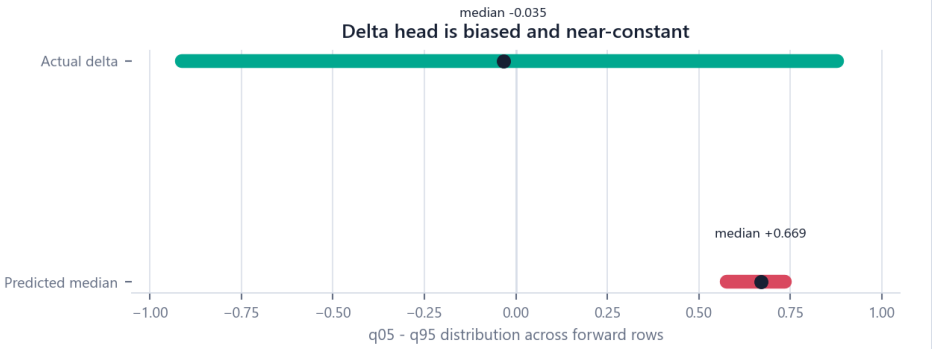
Forecasts are smooth, weak, and too uncertain to trade

Forward period: Jun 21-24 only. POC offsets are converted from ticks using the 0.1 USDT tick size.









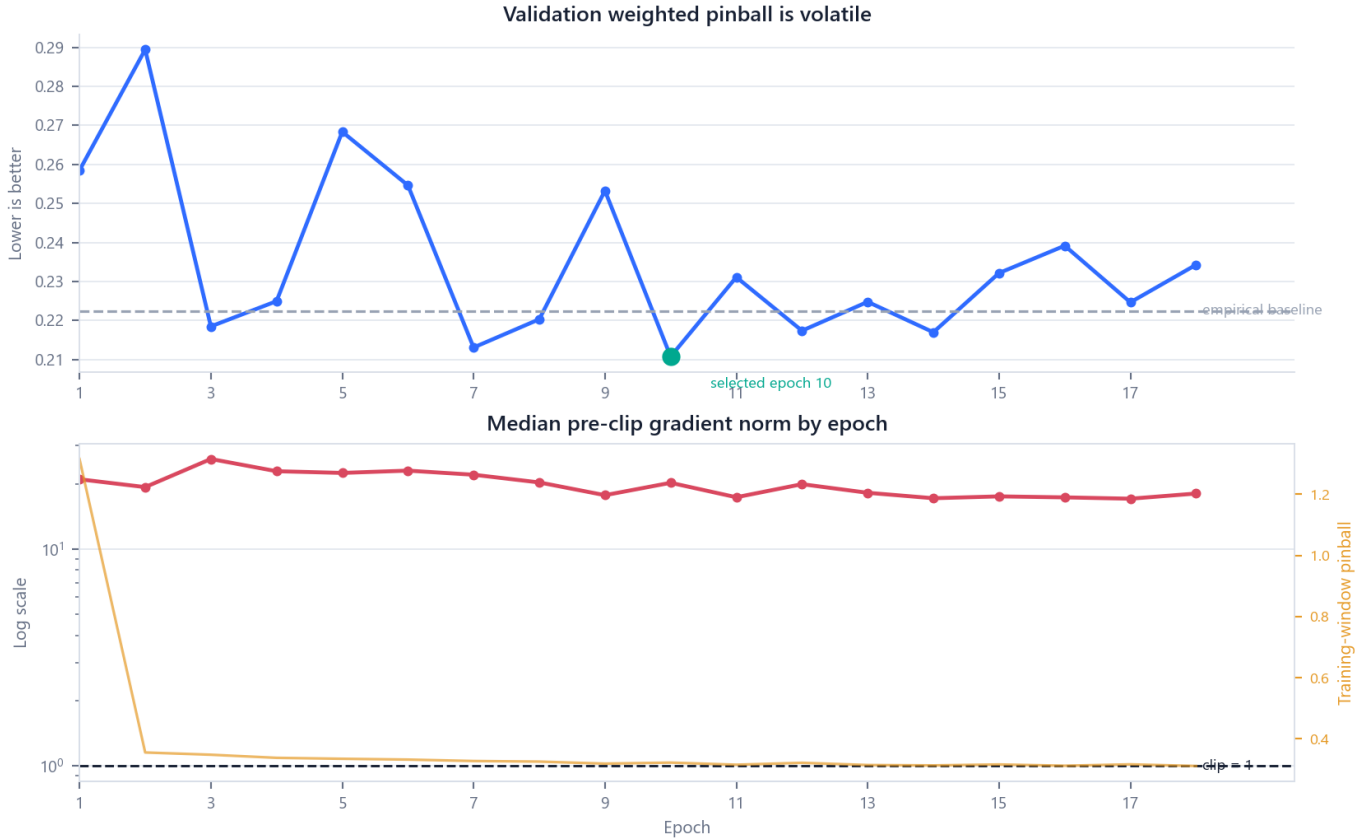
Five-minute diagnostics

POC Pearson / Spearman	**0.077 / 0.075**
Prediction std / actual std	**0.151**
Mean q10-q90 width	**190.47 USDT**
Intervals crossing zero	**100.0%**
Delta median bias / correlation	**+0.688 / -0.011**

Calibration coverage alone is not useful when every actionable interval includes zero.

Training selected a volatile checkpoint before schedules matured

Mechanics are proven; their causal effect requires controlled ablation.



100%

updates clipped

median norm 19.58 vs clip 1.0

36%

schedule completed

3,618 of 10,050 planned steps

0.25

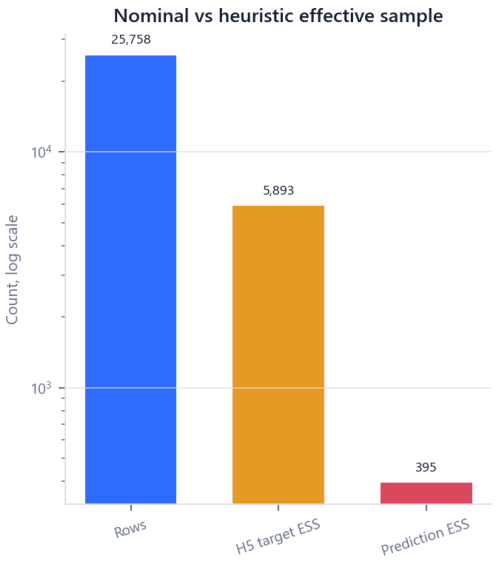
auxiliary weight

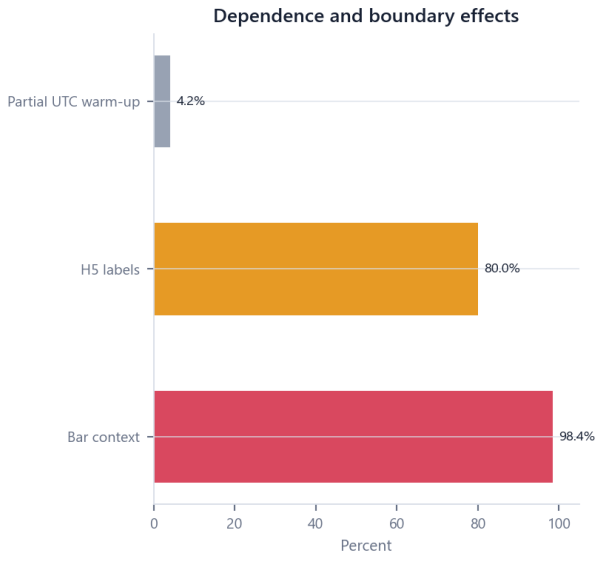
never decayed toward 0.05

Best epoch 10: 0.210765. Final epoch 18: 0.234276 (11.2% worse). Training-window pinball improved 0.321693 -> 0.310601. Final LR 2.35e-04; intended floor 3.0e-05.

25,758 rows are not 25,758 independent markets

Dependence, scale collapse, and narrow regimes reduce the information available to a 2.09M-parameter model.





Pipeline defects and validity limits

- 98.4% adjacent bar-context overlap; 80% H5-label overlap.
- Five future-minute reuses: 171,720 occurrences -> 34,440 unique minutes.
- L1 mid/bid/ask changes clip at about 22.84% each.
- UTC reset leaves 4.19% of rows with partial history.
- No independent receive time or native L2 sequence/checksum evidence.
- Event time substitutes for availability; decision lag is 0 ms.

18

training sessions

plus 2 selection / 4 forward

81.1

parameters per row

116,086 per session

1 seed

one fold trained

no variance estimate

4 days

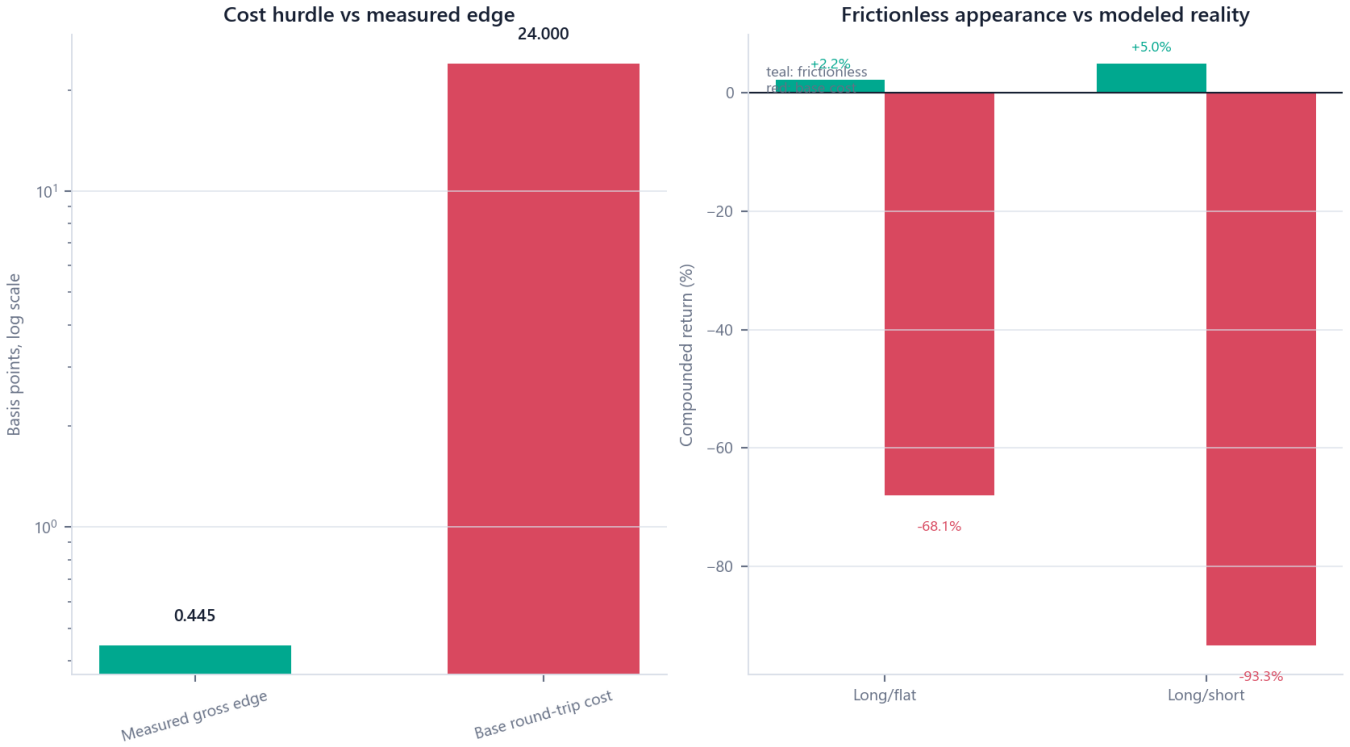
forward evidence

insufficient for regime/tail claims

Holdout integrity disclosure: Jun 25-30 target fields were inadvertently read during an auxiliary schema/count audit, not retained or used. Treat that period as process-accessed and replace it for final confirmation.

Transaction costs are not a sensitivity; they are the result

Five-minute non-overlapping execution proxy. Base cost is 12 bps per side, 24 bps round trip.



53.9x

cost / break-even

median long/flat

0.0475

direction MCC

accuracy 52.67%

0

primary trades

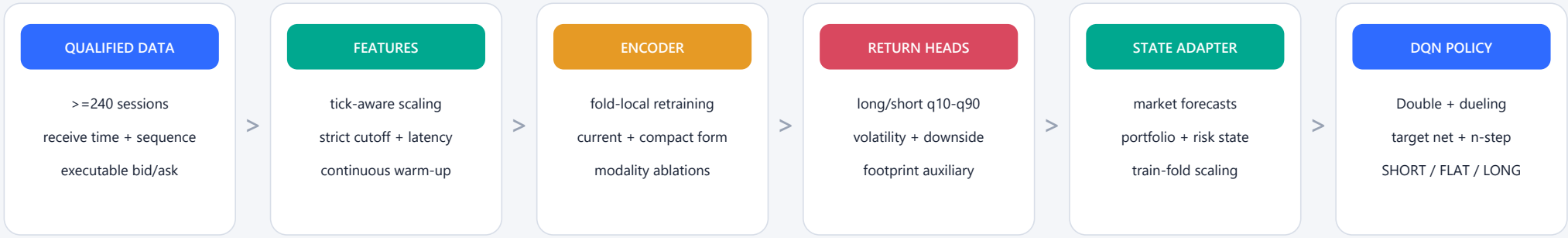
uncertainty-aware interval rule

Long/flat: 484 trades, frictionless +2.16%, base -68.07%, break-even 0.223 bps/side. Long/short: 1148 trades, frictionless +4.99%, base -93.34%.

Hybrid architecture: supervised signal first, DQN policy second

The policy stage stays locked until the encoder and a simple cost-aware policy pass every development gate.

HARD GATE >=5% proper-score gain + MCC >0.10 + non-negative simple policy after costs - otherwise stop before RL



Encoder protocol

RETRAIN	Old weights cannot seed a fold whose validation/test dates they saw.
COMPARE	Require simple baselines and a 0.2-0.5M compact encoder ablation.
FREEZE	Train the DQN head on fold-local or out-of-fold states first.
UNFREEZE	Only the top block, at lower LR, with a supervised-score guardrail.

Two endpoints - never conflate them

FORECAST	Proper score, MCC, calibration and interval width on rolling folds.
POLICY	Raw net PnL, drawdown, tail risk, turnover and paired confidence intervals.
CONTROL	DQN must beat the best frozen non-RL policy, not only random or buy-and-hold.
STATUS	Proposed architecture only - not implemented, validated, or approved for capital.

The environment is an accounting engine, not a reward toy

Use a fixed-small-notional exogenous replay: actions change portfolio state, never the recorded market path.

-1 SELL / SHORT

Target short; sell only the required position delta.

TARGET POSITION - NOT AN ORDER COMMAND

+0 FLAT

Target flat; close inventory or remain out of market.

TARGET POSITION - NOT AN ORDER COMMAND

+1 BUY / LONG

Target long; buy only the required position delta.

TARGET POSITION - NOT AN ORDER COMMAND

OBSERVE cutoff-safe state > **SELECT** target {-1,0,+1} > **WAIT** >=1m latency > **EXECUTE** bid/ask + costs > **MARK** liquidation equity

MARKET STATE

Transformer latent + return quantiles; spread, depth, imbalance, volatility, funding, time, liquidity, age and quality flags.

PORTFOLIO STATE

Current target, entry basis, unrealized PnL, holding time, equity, drawdown, turnover, prior cost, risk budget and episode time.

NET-EQUITY REWARD

$$r_t = \log(E_{t+1}/E_t) - \lambda_{dd} \cdot \max(0, DD_{t+1}-DD_t) - \lambda_{turn} \cdot |p_t-p_{t-1}|$$

Equity charges spread, fees, frozen slippage, funding and terminal close exactly once. Report raw net PnL and risk separately from shaped reward. Selecting the current target is hold/no trade; stale, OOD or risk-breach states force flat.

Train in stages; never joint-train from scratch on this corpus

Five rolling folds, fold-local states and replay, >=5 encoder seeds, >=10 DQN seeds for the finalist.

0 CERTIFY ENVIRONMENT

Causality, ledger, fills, forced close and deterministic replay.

Zero causal/accounting defects.

1 TRAIN ENCODER

Five rolling folds, fold-local states, >=5 seeds and simple baselines.

Forecast + simple-policy gates.

2 TRAIN FROZEN DQN

Freeze encoder; train adapter and DQN; >=10 finalist seeds.

Beats best frozen non-RL policy.

3 FINE-TUNE TOP BLOCK

Optional lower LR with a supervised forecast-score guardrail.

No regression or instability.

4 FREEZE POLICY

Freeze costs, latency, reward, risk, scaler and champion rule.

Immutable manifest + checkpoint.

5 SEALED HOLDOUT

Open one later >=60-session sealed period exactly once.

All gates pass; no retuning.

6 LIVE SHADOW

Frozen policy, no learning, capital or order route, for >=60 days.

Zero breaches; shadow gate passes.

INITIAL POLICY LEARNER

Dueling Double DQN + target network + Huber TD + day-balanced uniform trajectory replay + short n-step returns. QR-DQN or conservative Q regularization is an ablation only after the base learner and non-RL controls pass identical evidence.

CHRONOLOGY

Nested rolling train/select/test blocks with horizon-aware purge and embargo. Scalers, rewards, states and replay are fold-local. No validation or holdout transition enters replay.

Old checkpoint weights are disallowed wherever they have seen evaluation dates.

BASELINES + ABLATIONS

Always-flat, buy-and-hold, random, threshold, tree and DQN-without-transformer. Compare bars, L1, L2; compact vs current topology; frozen vs top-block fine-tune.

DQN must add paired raw net PnL over the strongest frozen non-RL policy.

NEGATIVE CONTROLS

Shuffle rewards and states within train days, randomize actions, verify regime failures, reconcile zero/base/stress costs and pre-register the trial budget plus one champion rule.

A shuffled control that appears profitable invalidates the experiment.

Promotion requires evidence at every layer

Thresholds are proposed now and must be registered before training; one failure stops the next stage.

A DATA
causal + executable
zero audit defects

B ENCODER
5 rolling folds
forecast gate

C SIMPLE POLICY
threshold / tree
net after costs

D DQN
paired incremental
beats non-RL

E CONFIRM
>=60 sealed sessions
one champion

F SHADOW
>=60 live days
no learning/capital

Go / no-go table

GATE	PASS CONDITION	EVIDENCE
Data	zero causal/accounting defects; <1% feature clipping	purge + embargo; fold-local state/replay
Forecast	>=5% paired proper-score gain; lower 95% CI >0	positive >=4/5 folds; MCC >0.10
Simple policy	non-negative raw net PnL after frozen base cost	must pass before DQN is trained
DQN value	paired incremental raw net-PnL lower 95% CI >0	versus best frozen non-RL policy
Economics	positive under stress; gross edge >=3x all-in cost	fills + latency + funding + close
Stability	>=80% positive seed-fold runs; >=4/5 fold medians	no fold/day supplies >30% of profit
Controls	shuffled states/rewards fail; leak sentinel rejected	all trials retained and registered
Confirmation	>=60 sealed sessions then >=60 shadow days	one champion; no retuning/learning

FAIL ANY DEVELOPMENT GATE -> STOP. DO NOT OPEN THE SEALED HOLDOUT.

Decision: proceed only to data and environment certification

The architecture is specified; its accuracy, incremental value and profitability remain unproven.

Proven from this run

- Objective is footprint prediction, not executable return.
- Median signal is weak; uncertainty crosses zero.
- Costs exceed measured edge by more than 50x.
- Targets are redundant/degenerate; sparse scaling loses magnitude.
- Training is clipping-dominated and stopped before schedule decay.

Not yet proven

- That DQN repairs accuracy or creates economic edge.
- That DQN beats the strongest frozen non-RL policy.
- That the transformer topology or L2 adds value.
- That counterfactual fills transfer at useful capacity.
- That any revised model is accurate or profitable.

Holdout integrity

An early schema/count projection traversed all 42,930 Arrow rows and physically read projected target fields in the Jun 25-30 shards. No sealed values were printed, retained, analyzed, or used. The period must nevertheless be treated as process-accessed and replaced for future confirmatory testing.

Action: retire Jun 25-30; physically seal a later >=60-session period with hashes and access logs.

Historical backtest outer_holdout_accessed=false remains true for artifact creation; it is not a current-session integrity claim.

Artifact lineage

DATASET	565c1d8e912ee426a1a92572d5be481ec5168d54d1ae53bd77fdd70d888e4df6
EXPERIMENT	25be2f446a75ad37120bba956ea86e46c0ecefa004ab3343d6745cdbee102fb3
CHECKPOINT	step-00002010
PLAN	chmt-q-transformer-dqn-research-v1

FINAL DECISION: RESEARCH PLAN ONLY

NOT IMPLEMENTED / NO DEPLOYMENT / NO CAPITAL